

The Data Layer

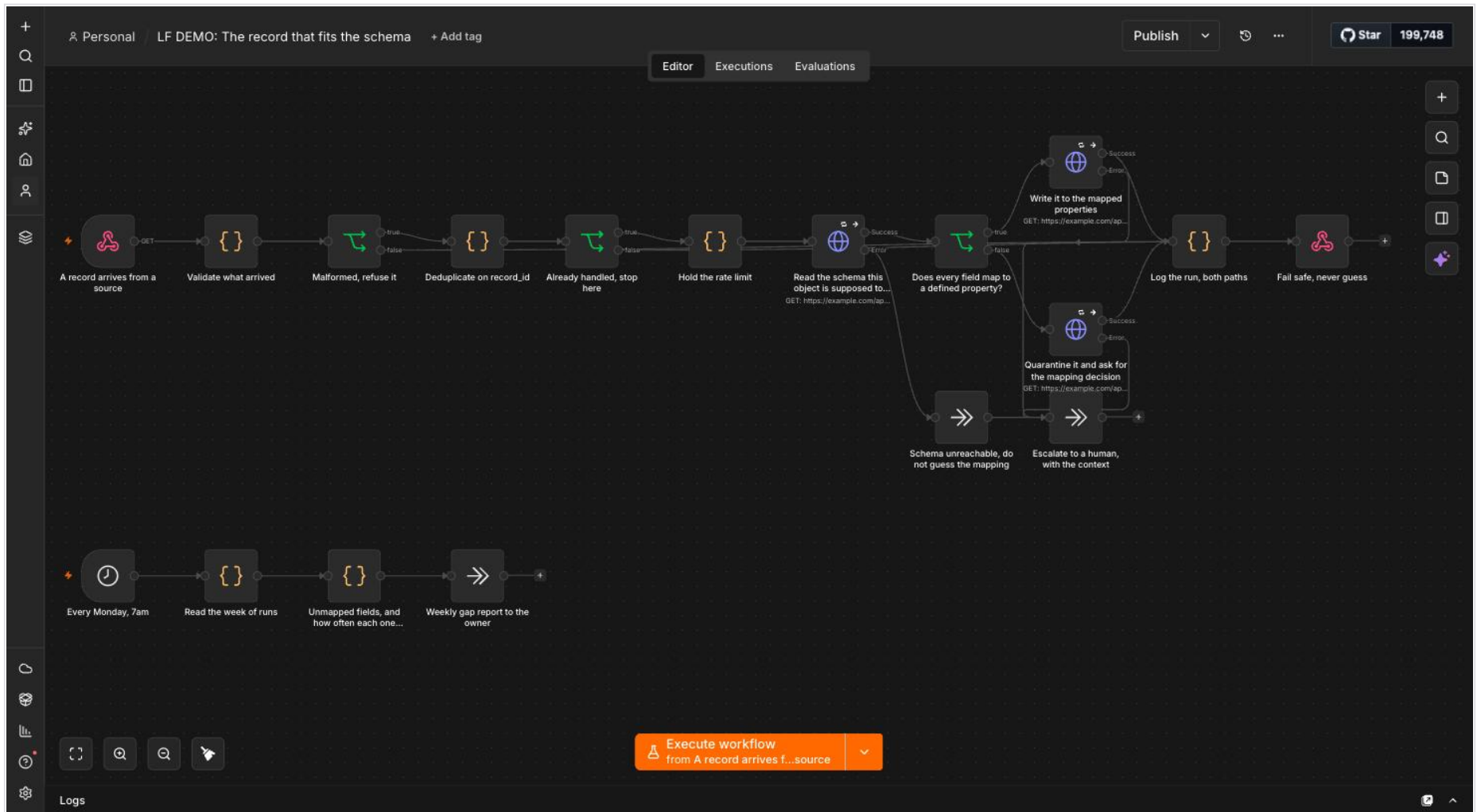
data-model-architecture · built in n8n

Most CRM problems are data-model problems wearing a workflow costume.

What it connects to: The schema has to hold for every system that writes into it. A property that only makes sense in one tool is the one that breaks the reporting later.

1 of 3 SYSTEM

Here's the system that enforces it — a record only gets written once every field maps to a property that exists.



2 of 3 THINKING

This is the gate that keeps the model clean: every field has to map to a property that exists, or the record is quarantined rather than guessed at. Guessing is how you end up with three fields meaning the same thing.

The screenshot shows a dark-themed user interface for a data processing tool. The main window is titled "Does every field map to a defined property?". The interface is divided into two main sections: "INPUT" on the left and "Parameters" / "Settings" on the right.

INPUT Section:

- At the top, it says "INPUT".
- Below that, there is a large area with a "No input data" message and a "Execute previous nodes" button.
- At the bottom, there is a "to view input data" link.

Parameters / Settings Section:

- There are two tabs: "Parameters" (selected) and "Settings".
- Under the "Parameters" tab, there is a "Conditions" section.
- The "Conditions" section contains a single condition: `{{ $json.fullyMapped === true && $json.objectExists === true }}`. To the right of the code is a checkbox labeled "is true" which is checked.
- Below the condition is a "+ Add condition" button.
- Below that is a toggle switch labeled "Convert types where required", which is currently turned off.
- Below the toggle is an "Options" section with a "+ Add option" button.
- At the top right of the "Parameters" tab, there is an orange button labeled "Execute step".

3 of 3 MORE

And the property architecture underneath it, with fill rates so you can see what's actually used.

Unmapped fields, and how often each one arrived

INPUT

→|

No input data

Execute previous nodes

to view input data

Parameters

Settings

Execute step

Mode

Run Once for All Items

Language

JavaScript

JavaScript

Code

Ask AI

```
1 // Unmapped fields, and how often each one arrived – the schema gap,
2 ranked.
3 const recs = Object.values($getWorkflowStaticData('global').records || {});
4 {});
5 const week = recs.filter(r => r.at > Date.now() - 7*864e5);
6 const counts = {};
7 for (const r of week) for (const f of (r.unmapped || [])) counts[f] =
8 (counts[f] || 0) + 1;
9 const ranked = Object.entries(counts).sort((a, b) => b[1] -
10 a[1]).slice(0, 5);
11 const quarantined = week.filter(r => r.quarantined);
12 return [{ json: {
13   received: week.length,
14   quarantined: quarantined.length,
15   worstFields: ranked,
16   note: ranked.length
17   ? `"${ranked[0][0]}" arrived ${ranked[0][1]} times with nowhere to
18   go. Decide the property once and ${quarantined.length} quarantined
```

Type \$ for a list of special vars/methods. Debug by using console.log() statements and viewing their output in the browser console.